Forecasting and Predictive Modeling – Sample Exam Subject

I. Select the correct answer(s) for the following questions. (10 x 0.3 = 3p).

1 When does it make sense to use the Drift model as baseline?

a) When the time series has seasons with equal variance.
b) When the time series is stationary.
c) When the time series has a trend.
d) When the time series has seasons where the variance changes with the level of the time series.

2. How many seasonal dummy variables are needed to encode the month of the observation in case of a regression model with an intercept term?

a) 1
b) 11
c) 12
d) 4

3. What do we use the ADF statistical test for?

a) To find the number of times differencing is needed to make a time series stationary.
b) To see if a series of autocorrelations is white noise.
c) To see if the variance in the seasonal component is changing with the level of the time series.
d) To see if a time series is stationary.

4. When should we consider using an ETS(A, Ad, N) model? Justify your answer.

a) When the time series is stationary.
b) When the time series has both seasons and trends.
c) When the time series has only trend.
d) When the time series has only seasons.

5. Which of the following measures should be maximized?

a) AIC
b) BIC
c) R-squared (R^2)
d) None of the above.

6. In the backshift notation what do we denote by B^2y_t ?

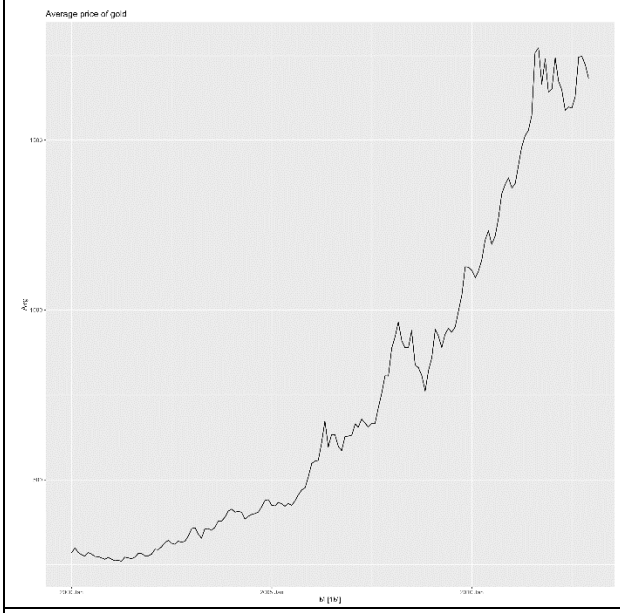
a) y_{t-2}
b) $(y_{t-1})^2$
c) $2 * (y_{t-1})$
d) $(Y_t)^2$

7. How do we determine the value of d for an ARIMA model?

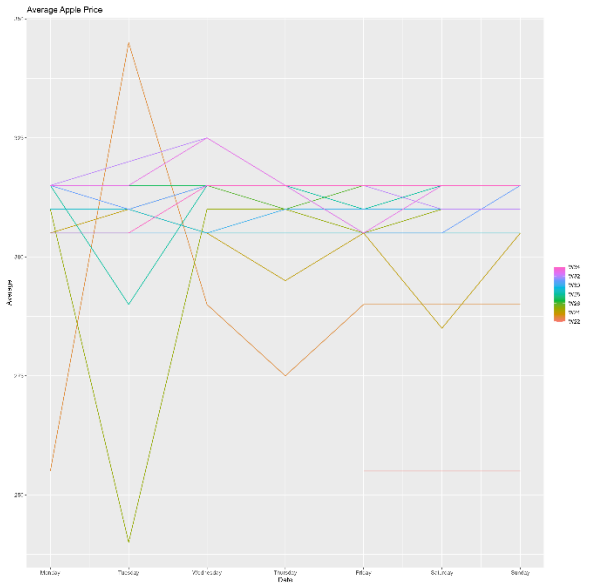
a) By looking at the ACF plot
b) By looking at the PACF plot
c) By using the KPSS or ADF statistical test
d) By looking at the time series plot.

II. For each plot below describe every piece of information that you can extract from it. (0.5 +0.5 + 1+1 = 3p)

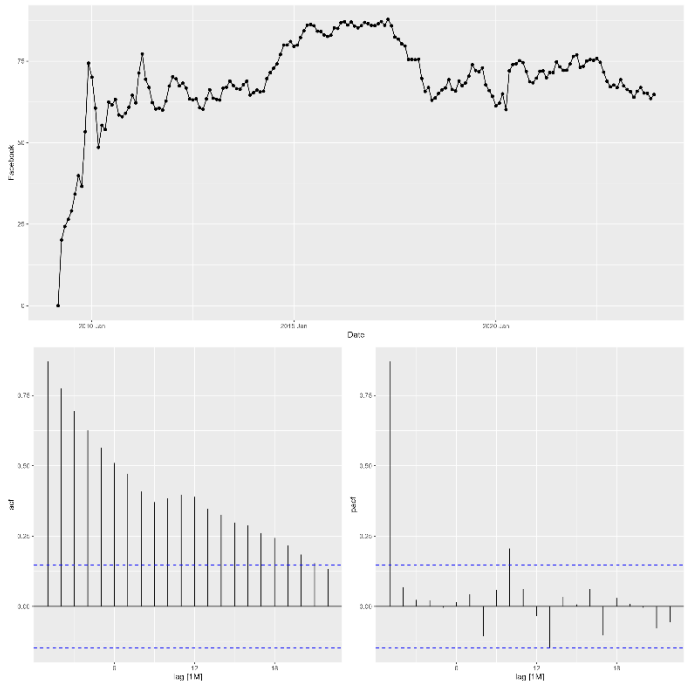
1.



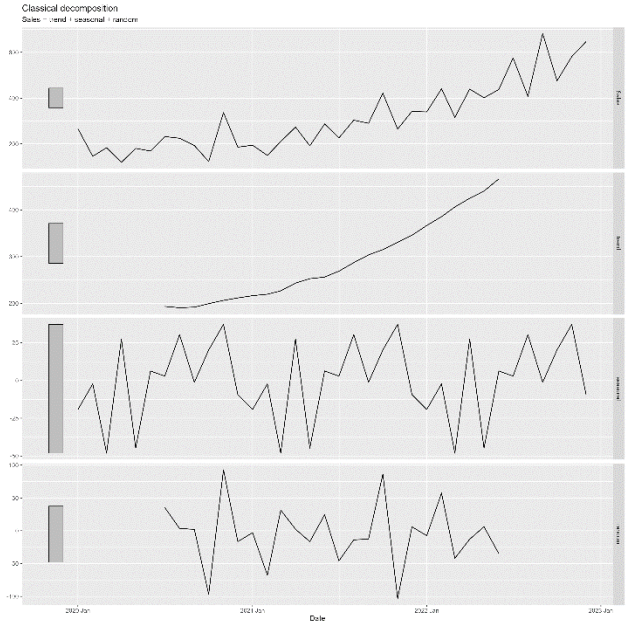
2.



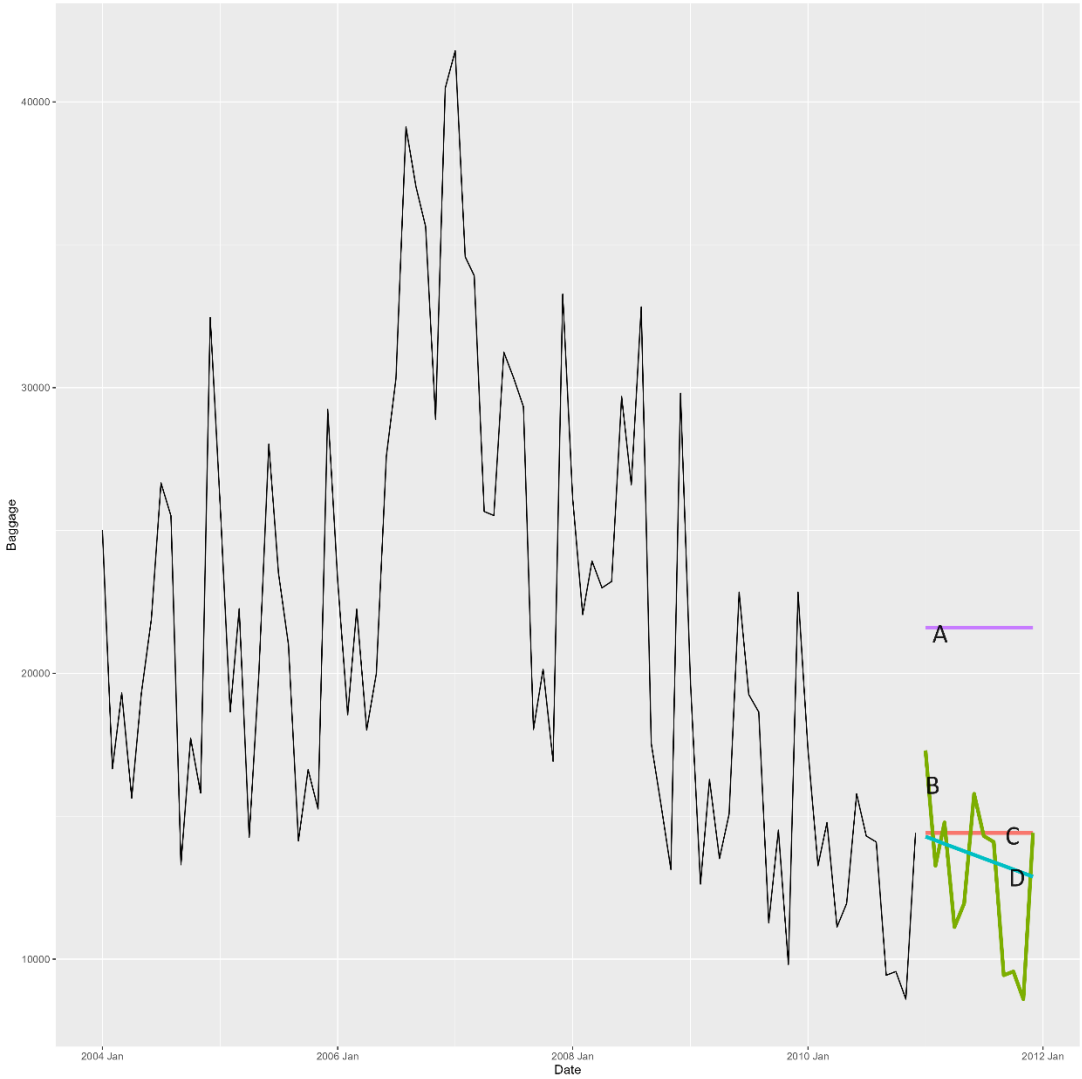
3.



4.



III. Below we have a time series about the number of baggage lost on flight of the United airline where we have used 4 different forecasting methods: Naïve model, Mean model, Seasonal Naïve and Drift Match each of these methods to the letters (A, B, C, D) on the plot of the forecasts. (1p)



IV. Consider the following time series made of 18 observations.

Day	*2024-01-01*	*2024-01-02*	*2024-01-03*	*2024-01-04*	*2024-01-05*	*2024-01-06*	*2024-01-07*	*2024-01-08*	*2024-01-09*
Value	*6*	*11*	*13*	*9*	*14*	*20*	*21*	*13*	*9*

Day	*2024-01-10*	*2024-01-11*	*2024-01-12*	*2024-01-13*	*2024-01-14*	*2024-01-15*	*2024-01-16*	*2024-01-17*	*2024-01-18*
Value	*16*	*12*	*22*	*14*	*13*	*5*	*14*	*11*	*19*

Answer the questions below (5 x 0.6 = 3p)

1. Draw a seasonal subseries plot of the values.

2. Compute the 3-MA (moving average) for the data. Do you think this is a good order for the given data?

3. Explain how would you add a dummy about weekend to the data.

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